Clone2vec captures continuous variation in clonal composition

To demonstrate the ability of clone2vec to capture clonal characteristics under different scenarios, we performed a series of simulations varying both the distribution of cells in the expression space, and the contribution of clones to different cell populations (see fig. 1E).

In the initial, simplest simulations, the cells were drawn from two distinct clusters in the expression space (fig. 1A). The clones were simulated to be of a fixed size (n=10), but varying in the proportion of their contribution to the two expression clusters (see fig. 1B). Clone2vec embedding of these clones effectively captures clone structure on the main axis of variance (see fig. 1C) with high correlation between the proportion of cells from cluster A (see fig. 1D).

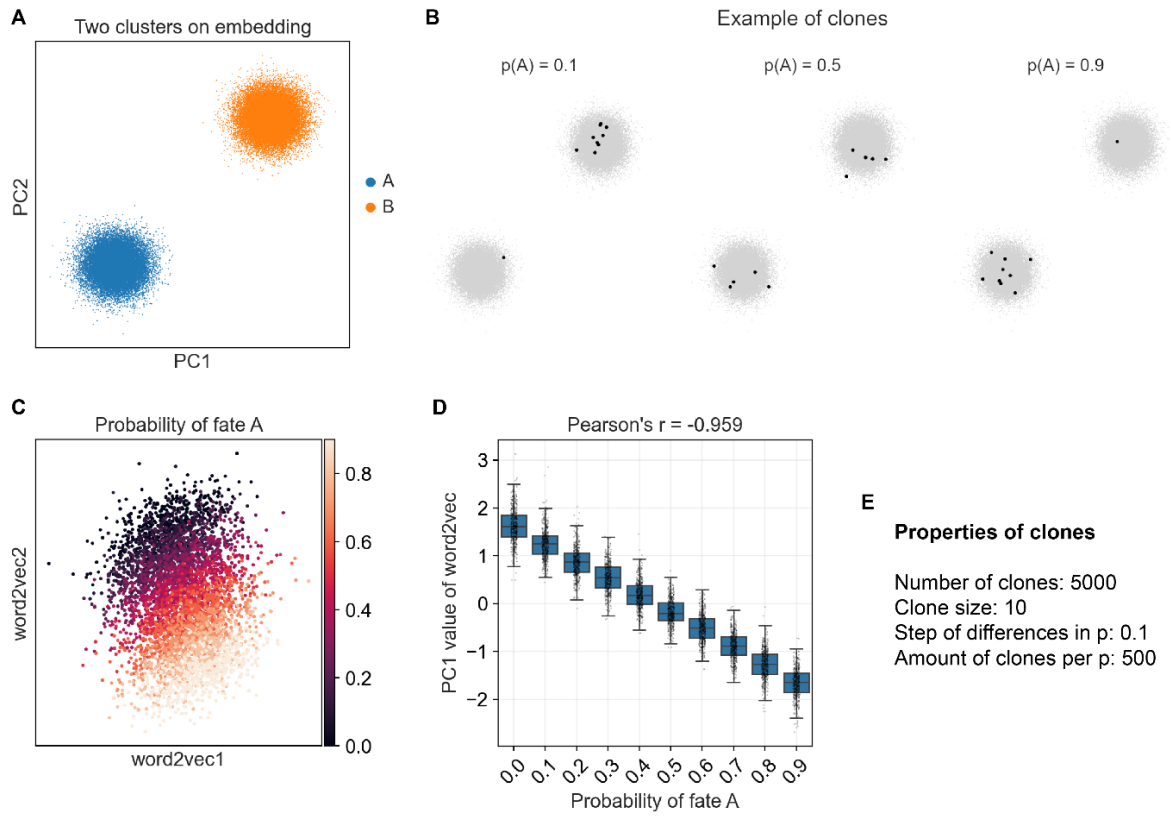

Figure 1
A. Expression positions of cells from the two generated expression clusters (A and B) are shown in the space of the first two expression PCs.
B. Examples of clones with different proportions of fates A and B.
C. Clone2vec embedding of clones, colored by the proportion of fate A.
D. Boxplot showing clone2vec2 embedding axis values for clones with different proportions of fate A.
E. Properties of the simulated clones.

Clone2vec captures discrete types of clones

Next, we examined the extent to which clone2vec is able to recover discrete clonal behaviors. To do so, we simulated multiple clones of a fixed size (n=15), each contributing cells to one of three expression populations (A,B,C - see fig. 2A) at a pre-defined proportion. The fate contribution of each clone is encoded in its name. Specifically, clone ABC contributes equally to all three expression states (5 cells each), while clone AAC contributes 10 cells to cell cluster A, and 5 to cell cluster C (fig. 2B).

Simulating a large number of such clones (fig. 2E), we find that they group together and create distinguishable clusters that are seen in both the space of the first two clone2vec axes (see fig. 2C), as well as UMAP embeddings based on the full clone2vec space (see fig. 2D), with almost perfect (ARI ≈ 1) match between the reconstructed clusters and the ground truth.

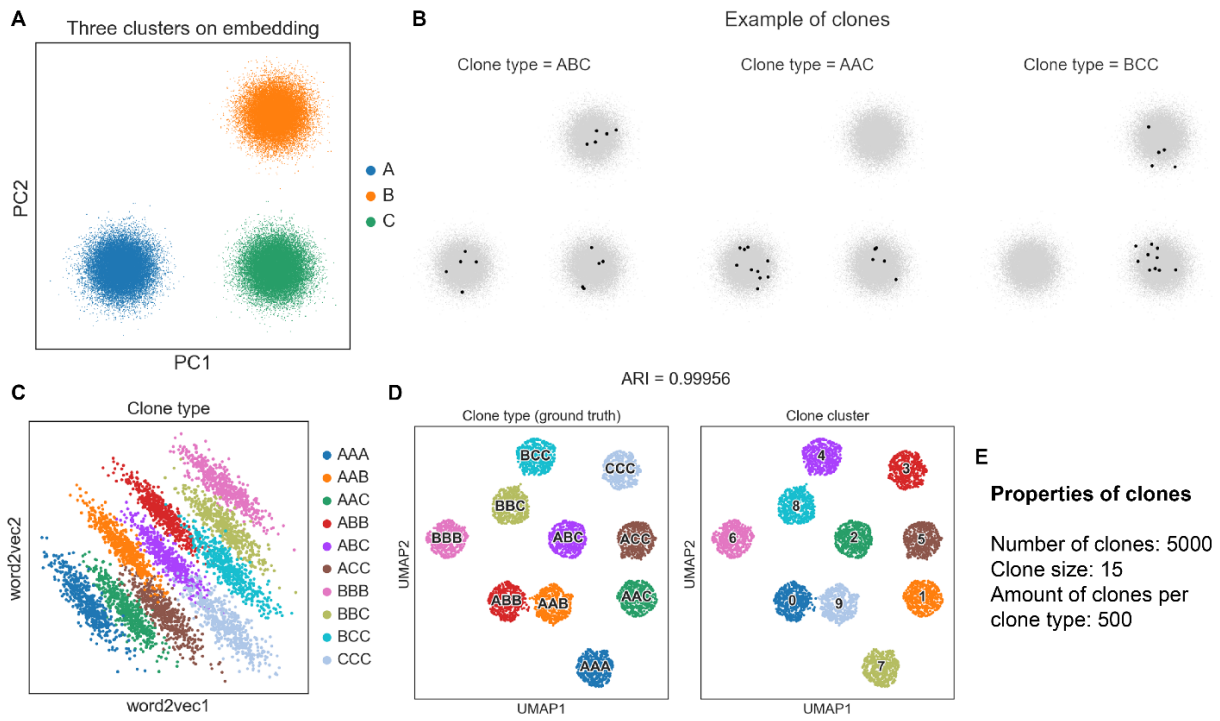

Figure 2
A. Generated expression embedding of cells with three distinct clusters — A, B, and C.
B. Examples of clones with different distributions across fates (the clones are named to reflect their contribution to each of the expression clusters).
C. Clone2vec embedding of clones shown on the first two axes.
D. UMAP embedding of the clone2vec representation of clones, colored by the ground truth (left), or the communities detected by the Leiden algorithm (right).
E. Properties of the simulated clones.

Clone2vec enables the identification of complex clonal behavior

The two scenarios simulated above could be studied with more conventional and simple methods. To demonstrate the ability of clone2vec to recover complex clone behaviors, we simulated a scenario where multiple clones contribute to complex, overlapping regions of the expression space. Specifically, we simulated a disc-like expression cell embedding (see fig. 3A), generating clones from overlapping domains as defined by different geometric patterns (see fig. 3B, and an example of clones in fig. 3C-D). The cross-like pattern of a clone (fig. 3C) simulates differentiation of four different fates from a central progenitor state. The incremental differential rotation of the cross-like pattern of clones is meant to simulate gradual biasing and continual nature of phenotypic differences of simulated terminal fates. In this way, the clones occupying the central part of this transcriptional embedding stay in a stem or a progenitor state (pink central dot, fig. 3A,B), whereas multipotent and progressively differentiating clones exiting a stem cell state are captured by patterns of concentric rings expanding to the periphery (with most differentiated cells occupying the outer ridge of the embedding, fig. 3B,D).

With moderate cluster sizes (n=30), clone2vec embedding clearly separated clonal clusters, perfectly matching the ground truth (see fig. 3E). This demonstrates the ability of clone2vec to to identify patterns of clonal distribution in cases of complicated gene expression distribution of clones, without clearly distinguishable clusters - a situation analogous, for example, to the mesenchymal differentiation trajectories.

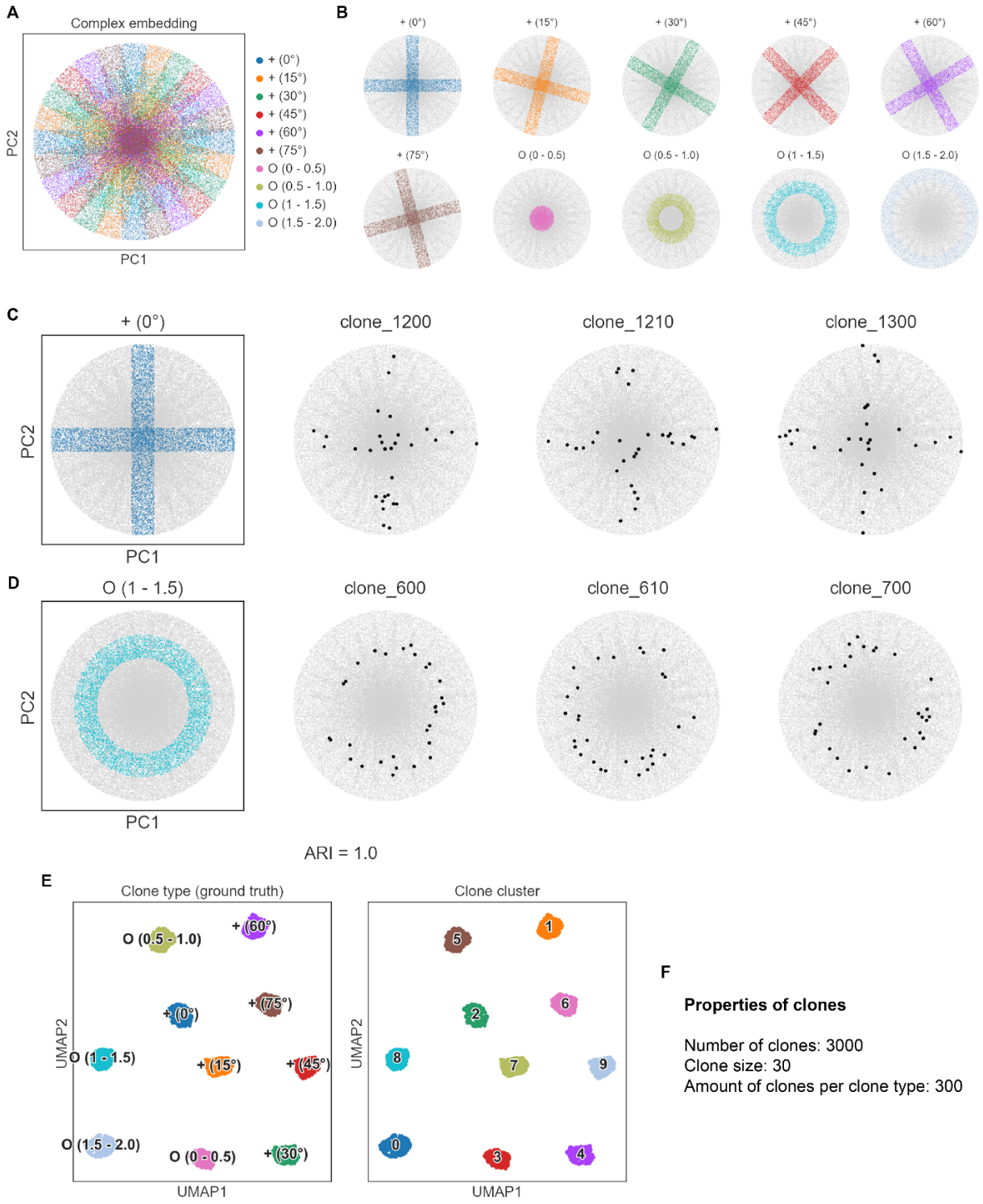

Figure 3

A. Generated gene expression embedding of cells, colored by clone types the cells belong to.

B. Illustration of the embedding domains occupied by different clone types.

C-D. Examples of clones with different behavior.

E. UMAP embedding of the clone2vec representation of clones, colored by the ground truth (left), or the communities detected by the Leiden algorithm (right).
F. Properties of the simulated clones.

Clone2vec is robust to subsampling and clonal size variation

Next, we evaluated the extent to which clone2vec reconstruction is robust under increasingly sparse sampling of clones. For this purpose, we subsampled the dataset simulated above to the 10% of its original size (see fig. 4A). This results in a much smaller clonal sizes (fig. 4D), and the simulated patterns are for most of the simulated clones the patterns become visually difficult to recognize (fig. 4B-C). Despite this, clone2vec was able to recover clonal structures with a good degree of accuracy (fig. 4E and 4J-M). Furthermore, although clones varied in size, clone2vec embedding was robust to this technical variation, separating clones based on the simulated fate patterns and not clonal size (see fig. 4F-G).

Finally, we compared *clone2vec* analysis with the more traditional approach of representing the clones in the space of cell types, where each clone is represented by a coordinate vector corresponding to the numbers of different cell types observed in the clone (see fig. 4H), followed by PCA dimensionality reduction. The embedding resulting from this approach yields relatively poor reconstruction of the original clusters (see fig. 4I), with the advantage of clone2vec becoming more apparent at sparser clonal coverage (fig. 4J).

Overall, our simulation and benchmarking results demonstrate that clone2vec is a robust and sensitive tool for identification of both discrete and continuous patterns of clonal fates, which can be used to explore complex clonal behaviors.

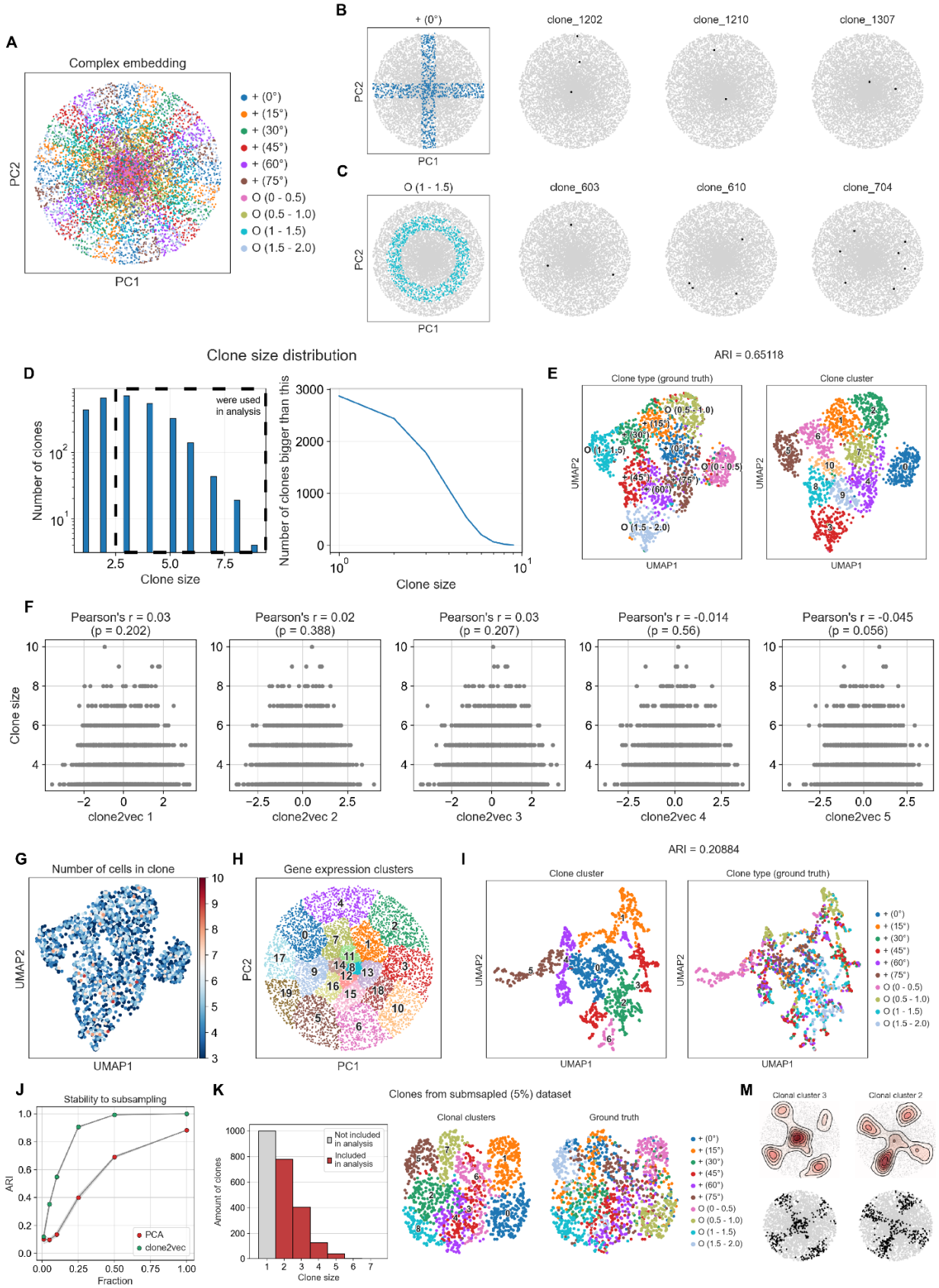
A
Complex embedding
PC2
PC1
+ (0°)
+ (15°)
+ (30°)
+ (45°)
+ (60°)
+ (75°)
O (0 - 0.5)
O (0.5 - 1.0)
O (1 - 1.5)
O (1.5 - 2.0)
B
+ (0°)
clone_1202
clone_1210
clone_1307
C
O (1 - 1.5)
clone_603
clone_610
clone_704
D
Clone size distribution
were used in analysis
Number of clones
Clone size
Number of clones bigger than this
E
ARI = 0.65118
Clone type (ground truth)
Clone cluster
UMAP1
UMAP2
F
Pearson's r = 0.03 (p = 0.202)
Pearson's r = 0.02 (p = 0.388)
Pearson's r = 0.03 (p = 0.207)
Pearson's r = -0.014 (p = 0.56)
Pearson's r = -0.045 (p = 0.056)
Clone size
clone2vec 1
clone2vec 2
clone2vec 3
clone2vec 4
clone2vec 5
G
Number of cells in clone
UMAP1
UMAP2
H
Gene expression clusters
PC1
PC2
I
ARI = 0.20884
Clone cluster
Clone type (ground truth)
J
Stability to subsampling
ARI
Fraction
PCA
clone2vec
K
Clones from submsapled (5%) dataset
Amount of clones
Clone size
Not included in analysis
Included in analysis
Clonal clusters
Ground truth
M
Clonal cluster 3
Clonal cluster 2

Figure 4

A. Gene expression cell embedding from Fig. 3A, subsampled to 10% of its original size.
B-C. Examples of clones with different behavior.
D. Distribution of the clone sizes after the subsampling procedure.
E. A UMAP embedding of clone2vec representation of clones, colored by ground truth (left), or by the communities detected by the Leiden algorithm (right).
F. Scatterplots showing the distribution of clone sizes along the first five clone2vec dimensions.
G. UMAP representation of clone2vec embedding of clones, colored by clone size.
H. UMAP representation of expression embedding of cells, colored by gene expression clusters.
I. Cell type distribution followed by UMAP embeddings based on the PCA coordinates of clones.
J. Scatterplot showing the stability of different approaches (clone2vec is green and PCA is red) to subsampling (in this case, all of the clones with size >= 2 were used in the analysis) based on Adjusted Rand Index (clones subsampled to a given fraction of cell number, compared to the total data).
K. UMAP representation of clone2vec embedding of clones under the 5% subsampling, colored by clusters (middle) or ground truth (right). A distribution of included clonal sizes is shown on the right.
M. Examples of different clone2vec-determined clonal clusters from the 5% subsampled dataset (M).